

Porter: Optimizing Delivery for the Digital Diner

Background:

Scenario: "Porter Delivery," a rising star in the food delivery industry, faces a significant challenge. Despite rapid growth and expansion, customer satisfaction scores related to delivery times have started to plummet in several key markets. The leadership team is under pressure to quickly identify the root causes and implement solutions to avoid losing market share to competitors. Trainees are brought in as data analysts to unravel this complex issue, armed with data and analytical tools. The scenario is practical, reflects real-world challenges, and emphasizes the importance of data analytics in strategic decision-making.

Problem Statement

The primary challenge is to analyze the factors influencing delivery times and identify areas for improvement to enhance customer satisfaction. This problem involves understanding delivery dynamics, store performance, and market conditions.

Stakeholders

- **Internal Stakeholders:** Operations team, marketing department, customer service, and the logistics team.
- **External Stakeholders:** Partner restaurants and delivery personnel.
- **Problem Definition:** To minimize delivery times while maintaining or improving service quality.
- **Data Requirements:** Detailed order data, including timestamps, store categories, order sizes, and delivery logistics.
- **Metric Development:** Metrics like average delivery time, on-time delivery rate, and peak times performance.
- **Insights & Actions:** Identifying bottlenecks, forecasting demand, and optimizing delivery routes.
- **Communication:** Presenting findings and actionable insights to stakeholders for implementation.

Data Link

<https://www.kaggle.com/datasets/ranitsarkar01/porter-delivery-time-estimation>

Data Dictionary

- Each row in this file corresponds to one unique delivery. Each column corresponds to a feature as explained below
- **market id:** integer id for the market where the restaurant lies
- **created at:** the timestamp at which the order was placed
- **actual delivery time:** the timestamp when the order was delivered
- **store_primary_category:** category for the restaurant

- order_protocol: integer code value for order protocol(how the order was placed ie: through porter, call to restaurant, pre-booked, third party etc)
- total_items_subtotal: the final price of the order
- num_distinct_items: the number of distinct items in the order
- actual_delivery_time: the timestamp when the order was delivered
- store_primary_category: category for the restaurant
- min_item_price: the price of the cheapest item in the order
- max_item_price: the price of the costliest item in order
- total_onshift_partners: number of delivery partners on duty at the time the order was placed
- total_busy_partners: number of delivery partners attending to other tasks
- total_outstanding_orders: total number of orders to be fulfilled at the moment

Initial Observations and Data Preprocessing Needs

- Missing Data: There are null values in market_id, actual_delivery_time, store_primary_category, order_protocol, total_onshift_partners, total_busy_partners, and total_outstanding_orders that need addressing.
- Data Types: The created_at and actual_delivery_time columns are in string format and should be converted to datetime for analysis involving time.
- Categorical Data: The store_primary_category is a categorical variable that may have a significant number of unique values, requiring examination and potential simplification.
- Potential for New Features: Calculating delivery duration, day of the week, and hour of the day from the timestamps could provide useful insights.

Data Cleaning and Preprocessing Guidance

To ensure the data is ready for analysis, the trainees should:

1. Handle missing values appropriately, considering imputation or removal based on the context.
2. Convert created_at and actual_delivery_time from string to datetime format for any time-based analysis.
3. Address any categorical data with a large number of unique values, possibly through aggregation or simplification.
4. Calculate additional features that could be relevant for analysis, such as delivery duration, day of the week, and hour of the day.

Basic-Level Questions

1. What is the distribution of order volumes across different markets?
 - Hint: Group orders by market_id and count the number of orders in each market.
 - Business Insight: Understanding order volume distribution can help identify high-demand markets, guiding resource allocation and marketing strategies.
2. How does the average delivery time vary across different store primary categories?

- Hint: First, calculate the delivery time for each order. Then, group the data by `store_primary_category` and calculate the average delivery time for each category.
 - Business Insight: This information can highlight which categories might be facing challenges in delivery efficiency, prompting targeted improvements or partnerships with stores.
3. What are the peak hours for order placements?
 - Hint: Extract the hour from `created_at` and analyze the frequency of orders across different hours.
 - Business Insight: Identifying peak ordering times can assist in staffing optimization, ensuring enough delivery partners are available, thus reducing wait times and improving customer satisfaction.
 4. Which days of the week see the highest order volumes?
 - Hint: Convert `created_at` to datetime format, extract the day of the week, and count the number of orders for each day.
 - Business Insight: This analysis can guide promotional activities and special offers to either capitalize on high-demand days or boost orders on slower days.
 5. What is the correlation between the number of items in an order and the subtotal?
 - Hint: Use a correlation function to explore the relationship between `total_items` and `subtotal`.
 - Business Insight: Understanding this relationship can inform menu pricing strategies and promotions to increase order value.
 6. Analyze the distribution of delivery times. Are there notable outliers?
 - Hint: After calculating delivery durations, use statistical methods or visualization to identify outliers.
 - Business Insight: Identifying outliers can help pinpoint exceptional delays or inefficiencies in the delivery process, leading to targeted improvements.
 7. What percentage of orders are delivered within 30 minutes?
 - Hint: Calculate delivery time for each order and then determine the proportion that meets the 30-minute criteria.
 - Business Insight: A high percentage of timely deliveries can be a strong selling point for customer satisfaction and retention.
 8. How do order volumes correlate with the number of busy partners?
 - Hint: Analyze the relationship between order volume and `total_busy_partners` to find patterns or correlations.
 - Business Insight: This can help in optimizing partner schedules and availability, ensuring delivery efficiency is maximized during high-order periods.
 9. What is the impact of the day of the week on average delivery time?
 - Hint: Group data by the day of the week and calculate the average delivery time for each group.
 - Business Insight: Insights into how delivery times vary by day can inform operational adjustments and customer expectations management.
 10. Identify the top 5 stores with the longest average delivery times.
 - Hint: Calculate average delivery times per store and then sort to find the stores with the longest times.

- **Business Insight:** This could indicate potential issues with specific stores or areas, allowing for targeted operational improvements or discussions with store partners.

Medium-Level Questions

Evaluate the efficiency of different order protocols in terms of delivery time.

- **Hint:** Group the data by `order_protocol` and calculate the average delivery time for each protocol.
- **Business Insight:** Understanding which protocols yield faster deliveries can guide training and technology investments, enhancing overall efficiency.

How does the average order value vary by market?

- **Hint:** Calculate the average subtotal for orders in each `market_id`.
- **Business Insight:** Insights into market-specific spending patterns can inform targeted marketing strategies and menu adjustments to increase revenue.

What is the relationship between the number of on-shift partners and delivery times during peak hours?

- **Hint:** Identify peak hours first, then analyze how delivery times correlate with the number of `total_onshift_partners` during these times.
- **Business Insight:** This analysis can help in optimizing partner schedules to ensure sufficient coverage during peak times, potentially reducing delivery times.

Determine the impact of total items and distinct items in an order on delivery time.

- **Hint:** Use regression analysis to explore how `total_items` and `num_distinct_items` affect delivery time.
- **Business Insight:** Understanding how order complexity impacts delivery time can aid in setting realistic delivery expectations and improving customer satisfaction.

Identify trends in delivery time fluctuations throughout the week.

- **Hint:** Calculate average delivery times for each day of the week and compare.
- **Business Insight:** Identifying specific days with longer delivery times can help in planning interventions or incentives to improve efficiency and customer experience.

Assess the role of item price range in an order's total value.

- **Hint:** Calculate the price range (difference between `max_item_price` and `min_item_price`) for each order and correlate it with subtotal.
- **Business Insight:** This can reveal if a wider range of item prices in orders correlates with higher order values, potentially guiding menu pricing strategies.

What effect does store primary category have on the likelihood of orders being above average value?

- **Hint:** First, determine the average order value (subtotal), then calculate the proportion of orders above this average within each `store_primary_category`.
- **Business Insight:** Insights into high-value categories can direct promotional efforts and partnerships to maximize revenue.

Explore the distribution of delivery times. Is there a significant variance within markets or categories?

- Hint: Calculate the standard deviation of delivery times by market and category.
- Business Insight: High variance might indicate inconsistent service levels, highlighting areas for operational improvement.

How does the availability of delivery partners (busy vs. on-shift) affect delivery times in different markets?

- Hint: Analyze the correlation between the ratios of busy/on-shift partners and average delivery times across markets.
- Business Insight: This can highlight markets where partner availability mismatches demand, guiding recruitment or scheduling adjustments.

Compare first-time and repeat customer order patterns in terms of items and subtotal.

- Hint: You might need to infer or create a flag for first-time versus repeat customers based on order frequency by customer ID, then compare the average total_items and subtotal.
- Business Insight: Understanding the differences in ordering patterns can help tailor marketing and loyalty strategies to encourage higher spend and repeat business.